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COMPUTER SECURITY WK6

1.SCENARIO ANALYSIS:

Authentication Scenario: Banking App Login

A customer uses a mobile banking application to access their bank account and perform transactions. When the customer opens the application, they enter their username and password. The application sends the login information securely to the bank's authentication server for verification. The server checks the credentials against the customer's registered account details. If the information is correct, the customer is authenticated and allowed to access their account. For additional security, the bank may require multi-factor authentication, such as a one-time password (OTP) sent by SMS or generated by an authentication application. If the customer enters incorrect credentials several times, the account may be temporarily locked to prevent unauthorized access. The banking application should also use encryption to protect sensitive information during communication. This authentication process helps ensure that only authorized customers can access accounts, reducing the risk of identity theft, fraud, and unauthorized transactions.

2.CONCEPT RESEARCH:

Multi-Factor Authentication (MFA)

Multi-Factor Authentication (MFA) is a security method that requires users to provide two or more forms of verification before gaining access to an account or system. These factors usually include something the user **knows**, such as a password or PIN; something the user **has**, such as a mobile phone or security token; and something the user **is**, such as a fingerprint or facial recognition. MFA provides stronger security than using a password alone because an attacker who obtains one authentication factor may still be unable to access the account. It is widely used in banking, email, cloud services, and business systems. MFA helps reduce unauthorized access, account takeover, identity theft, and other security threats.

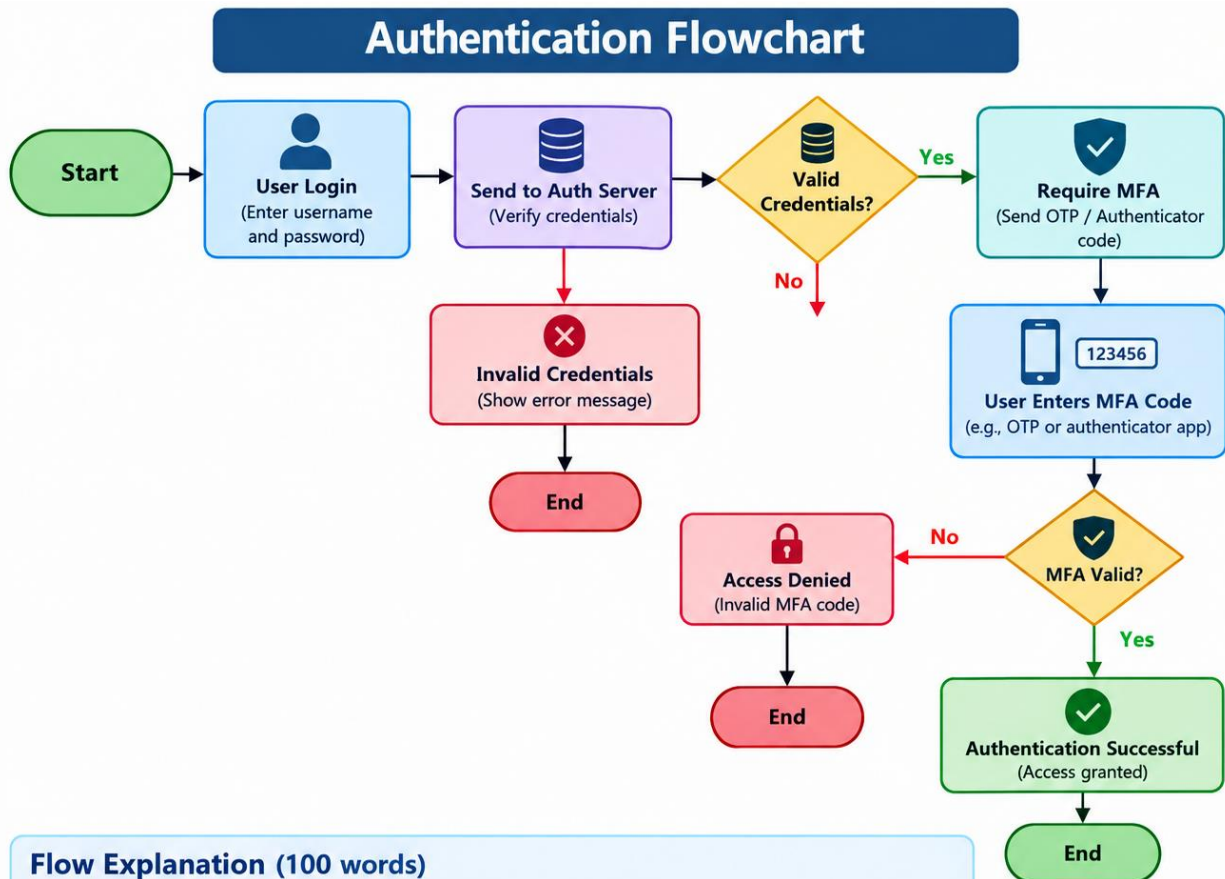
3.TOOL PRACTICE:

Tool Practice: Setting Up MFA in Auth0

I practiced setting up Multi-Factor Authentication (MFA) using Auth0. First, I accessed the Auth0 Dashboard and opened the **Security** section, where I selected **Multi-factor Authentication**. I enabled MFA and configured an authentication method, such as an authenticator application. During login, the user was required to enter their normal username and password and then provide an additional verification code generated by the authenticator app. This practical exercise demonstrated how MFA adds an extra layer of security to user accounts. Even if a password is stolen, an attacker would still need the second authentication factor. The practice also showed that MFA can help protect applications from unauthorized access, account

takeover, and identity theft. Overall, Auth0 makes it possible to strengthen authentication by combining passwords with additional verification methods.

4.DIAGRAM:



Flow Explanation (100 words)

1. The user enters their username and password in the application.
2. The application sends the credentials to the authentication server for verification.
3. If the credentials are valid, the user is prompted for a second factor (MFA), such as an OTP or authenticator code.
4. The user enters the MFA code, which is verified by the server.
5. If the MFA code is valid, the user gains access to the application. If the code is invalid or the credentials are incorrect, access is denied and an error message is shown. This process adds an extra layer of security and helps prevent unauthorized access.